

Enhancing Digital Education in Rural Communities

Digital and Hybrid Teacher Professional Development in Rural Sabah, Malaysia

Assignment 2

Group 3

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1. Use Case, Primary Stakeholder, and Strategic Goal

1.1 Use case and geographic focus

The use case is **digital and hybrid teacher training in rural Malaysia under low-resource conditions**. We focus on the **Interior Division of Sabah** (Keningau, Tambunan, Tenom, Nabawan, and Sipitang) and nearby rural districts (Pitas, Tongod, and Ranau). Three facts make this a good case: eight of the ten poorest districts in Malaysia are in Sabah; the Sabah State Minister of Education said in December 2025 that **300 to 400 rural Sabah schools still have no internet** (The Star, 2025) and a 2025 UMS study found that rural Sabah teachers face weak infrastructure, heavy workloads, and professional isolation (Jaafar et al., 2025).

1.2 Primary stakeholder

We assume the primary stakeholder is the **Bahagian Profesionalisme Guru (BPG)**, the unit inside the Ministry of Education Malaysia (MOE) in charge of teacher professional development. It is led by its Director, **Datuk Dr. Mistirine binti Radin**. BPG sets training policy, plans CPD programmes, checks that in-service training meets national standards, and evaluates programmes yearly (MOE, n.d.-a). It sits in the same sector as the Institut Pendidikan Guru Malaysia (IPGM) and the Institut Aminuddin Baki (IAB) (E-Bantuan Malaysia, 2025). BPG is the right stakeholder here because it owns teacher training and the main programmes behind it, such as PPGB 2.0 and SGM 2.0 (MOE, 2023; Slideshare, 2024).

1.3 Strategic goal

From BPG's point of view, the strategic goal is to **improve training for rural teachers in Malaysia, with a focus on Sabah's Interior Division, so teachers can grow their digital teaching skills in real rural conditions. The goal aligns with three national plans: the Dasar Pendidikan Digital (DPD) 2023–2030, the Thirteenth Malaysia Plan (RMK-13, 2026–2030), and the Standard Guru Malaysia 2.0.**

1.4 Where the goal comes from

The goal reflects both national policy and the real rural problems from the source material: weak internet, not enough devices, and poor follow-up after training. We assume four pressures on BPG are driving it.

First, **national policy**. The DPD 2023–2030, launched by Minister Fadhlina Sidek on 28 November 2023, commits MOE to producing "digitally fluent pupils." It reports that only **2.2% of Malaysian teachers are at advanced digital level**

and 39.9% at intermediate, with rural areas worse off (MOE, 2023). RMK-13, tabled on 31 July 2025, keeps a "Raise the Floor" pillar for underserved regions. Together, these plans make rural teacher training a measurable target.

Second, **a teacher shortage**. On 9 April 2025, MOE launched the *Pelan Tindakan Nasional Pendidikan 2025–2030* to handle a shortage of almost 10,000 teachers in Maths, English, and science, mostly in Sabah, Sarawak, Kelantan, and Terengganu. MOE plans to recruit 12,000 new teachers from July 2025, plus a six-month intensive programme (PLIG) for new graduates (PrezCom, 2025). Many will be posted to rural schools, making PPGB 2.0 and its follow-up support very important.

Third, **a change in how training is tracked**. On 30 December 2025, Minister Fadhlina Sidek announced that the central training tracking system (SPLKPM) will be shut down in 2026, to "place full trust" in teachers (Utusan Malaysia, 2025). Removing this system just as DPD 2023–2030 and RMK-13 start means any new solution needs its own way to track training.

Fourth, **state-level plans match**. JPN Sabah, under Director Raisin bin Saidin, runs the *Sepakat Sabah Hebat (SSH) 2023–2025* plan. Its four pillars mirror the national *Program Transformasi Sekolah 2025 (TS25)*: Effective Leaders, **Quality Teachers**, PTA and Community Support, and Meaningful Learning Environment (JPN Sabah, 2022). The "Quality Teachers" pillar matches any BPG plan for rural digital training, making Sabah a natural fit.

2. Stakeholders

The strategic goal involves groups that run the work, groups that influence it, and groups affected by the result. We list them below with specific names.

Federal policy and training governance. *Bahagian Profesionalisme Guru (BPG, MOE)* under Datuk Dr Mistirine binti Radin owns the teacher training policy and the SGM 2.0 standard. *Bahagian Sumber dan Teknologi Pendidikan (BSTP, MOE)* owns the DELIMA 2.0 learning platform and runs the *Pusat Kegiatan Guru (PKG)* teacher centres. *Institut Pendidikan Guru Malaysia (IPGM)* trains new teachers through its campuses, including **IPG Kampus Keningau** inside the Interior Division. *Institut Aminuddin Baki (IAB) Cawangan Sabah* in Kota Kinabalu trains school leaders across Sabah and WP Labuan (IAB, 2025).

State and district. *Jabatan Pendidikan Negeri Sabah (JPN Sabah)* under Raisin bin Saidin runs the SSH 2023–2025 plan. Each *Pejabat Pendidikan Daerah (PPD)* in the Interior (Keningau, Tambunan, Tenom, Nabawan, Sipitang, Pitas) picks schools, tracks progress, and sends support. At school level, **Guru Besar** (headmaster) and **Penolong Kanan** (senior

assistant) control teacher time and decide who joins training.

Teachers and mentors. *Rural teachers* in Sabah are the main users whose skills and classroom work should improve. They receive the *Bayaran Insentif Mengikut Lokasi dan Tahap Kesusahan (BIMLTK)* allowance of RM500/RM1,000/RM1,500 per month for Rural 1/2/3 schools (Free Malaysia Today, 2025). *School digital leads and mentor teachers* support peer learning — most notably through MDEC's **#CikguJuaraDigital**, where 30 Champion Teachers in Sabah have trained 1,434 others since 2020 (Sabah Media, 2024).

Internet and community infrastructure. *Suruhanjaya Komunikasi dan Multimedia Malaysia (MCMC)* runs the **131 NADI centres in Sabah**. In July 2025, it launched the *NADI Ambassador Sabah* pilot, its first programme to list teachers as target users (Borneo Post, 2025a). MCMC also runs *JENDELA Phase 2*, which got RM1.4 billion in November 2025 for 800 new sites in Sabah over five years (Borneo Post, 2025b).

Industry, foundations, and UN partners. *Malaysia Digital Economy Corporation (MDEC)* runs **MyDigitalMaker** and its teacher arm **#CikguJuaraDigital**, reaching about 80,000 students across 800 Sabah schools by mid-2024 (Sabah Media, 2024). *Yayasan PETRONAS* runs **Program Duta Guru** (training 4,500 STEM teachers by 2030) and the *CERDIK* device programme. *UNICEF Malaysia* runs the **Future Skills for All (FS4A)** platform, which has both online and offline modes and reached 43,838 children and 3,721 teachers in 2024 (UNICEF Malaysia, 2024). *CelcomDigi* and *Maxis* (eKelas, added to DELIMa in 2023) are the main telco partners.

Academic partners. *Universiti Malaysia Sabah (UMS)* — mainly its Faculty of Education and the *Unit for Rural Education Development (UPPLB)* — provides peer-reviewed research on rural teacher conditions (Jaafar et al., 2025).

Community and parents. Village governance runs through the *Ketua Kampung* (village head), the *JKKK* village committee, and the *Ketua Anak Negeri* from the Native Court system. They affect teacher attendance and after-hours access to training sites. *Parent-Teacher Associations (PIBG)* link the school to the community. For local-language content (e.g. Kadazandusun), the *Kadazandusun Cultural Association (KDCA)*, led by Tan Sri Joseph Pairin Kitingan (2025–2028 term), is the community authority.

Students in rural schools are the final beneficiaries — their learning depends on whether teachers can run low-data, practical lessons.

3. Project Goals and ContextSquaREs

From the strategic goal in Section 1, we identify three project goals, one from each of the three stakeholder views: the teacher, the school leader, and the people who track results at the district and national levels.

Project Goal 1 (teacher view). Rural teachers should be able to get and use practical training even when the internet and devices are limited.

Project Goal 2 (school leader view). Schools should support teacher learning through simple peer learning, mentoring, and coaching.

Project Goal 3 (tracking view). District and national offices should be able to track, recognise, and scale up rural teacher training based on real evidence.

3.1 ContextSquaRE 1 — Project Goal 1

Project goal. Rural teachers should be able to get and use practical training even when the internet and devices are limited.

Stakeholders. Rural teachers in Sabah Interior, Guru Besar, rural pupils, district coaches from JPN Sabah and the PPDs, BPG as national policy owner, BSTP as owner of DELIMa 2.0, and MCMC NADI centres as community infrastructure.

1. Key Challenges. Teachers in the Interior Division deal with weak 4G, few devices, and pupils with different resources at home. As of December 2025, 300 to 400 rural Sabah schools still have no internet. Time is tight: a new teacher at a Rural 2 school has extra paperwork, drives three or more hours to the nearest PKG teacher centre, and has little time for live online training. Professional isolation is a known problem (Jaafar et al., 2025). Teachers usually stay only two to three years before asking for a transfer, which makes long training investments less useful.

2. Existing Solutions. Malaysia already has useful pieces in place. DELIMa 2.0 includes a professional development module, short certified micro-courses, a teacher knowledge-sharing space, and single sign-on to Apple, Microsoft, and Google learning tools (BSTP, 2025). PPGB 2.0 gives new teachers a structured induction and mentoring programme (Slideshare, 2024). MDEC's **#CikguJuaraDigital** has 30 Champion Teachers in Sabah who have trained 1,434 others since 2020 (Sabah Media, 2024). The 131 MCMC NADI centres have internet and training space, and the *NADI Ambassador Sabah* pilot (24 July 2025) reached teachers for the first time (Borneo Post, 2025a). These structures provide a policy and platform base rather than a blank starting point.

3. Shortcomings. Existing support is scattered and not well adapted to rural conditions. Getting training does not always mean teachers apply it in the classroom, especially when the training is generic or needs a stable connection. DELIMa 2.0 assumes the

user is always online; a 2022 study in Sibu already showed similar teacher-adoption and connectivity problems (Keling et al., 2025). DELIMa has no Kadazandusun or Iban content for teachers. BPG has not published a mapping from SGM 2.0 to UNESCO ICT-CFT (UNESCO, 2018), which makes rigorous curriculum design harder. PKGs and IAB are too far to reach in a normal workday, and NADI centres are not officially linked to BPG's in-service training system.

4. Proposed Solution Direction. A digital solution should provide mobile-friendly, low-bandwidth, offline-first training content, ready-to-use classroom templates, and asynchronous evidence submission, written against SGM 2.0 and DPD 2023–2030 skill levels. It should work on cheap Android devices and sync whenever a connection is available — over 4G, NADI Wi-Fi, or USB sync during a PPD visit. Content should be built with #CikguJuaraDigital Champion Teachers and include at least Bahasa Malaysia and Kadazandusun for relevant modules. Training done should count toward PPB credits under SGM 2.0, so teachers keep their progress when they move schools.

3.2 ContextSquaRE 2 — Project Goal 2

Project goal. Schools should support teacher learning through simple peer learning, mentoring, and coaching.

Stakeholders. Guru Besar and Penolong Kanan at rural schools; IAB Cawangan Sabah for school leader training; BPG as national policy owner; JPN Sabah and the PPDs; mentor teachers including #CikguJuaraDigital; the PKG teacher centres in each Interior district; and pupils who benefit from better teaching.

1. Key Challenges. Interior schools are small; a typical rural primary school has 4 to 12 teachers, which makes it hard to run a full peer learning group inside one school. Guru Besar are often new leaders themselves, with little protected time for mentoring. Schools in the same cluster can be three or more hours apart on unpaved roads, and high turnover breaks up any peer learning that starts. When peer support happens, it is usually on WhatsApp or Telegram: not saved, not official, not recognised. Rural timetables have no dedicated time for peer learning.

2. Existing Solutions. IAB Cawangan Sabah runs structured training for school leaders (including the NPQEL pathway) for Sabah and WP Labuan (IAB, 2025). The PKG network has 367 centres nationwide and is the only MOE training site physically present in every Interior district (PKG KPM, 2025). SSH 2023–2025 lists “Quality Teachers” as one of four pillars, giving JPN Sabah a mandate for peer learning work (JPN Sabah, 2022). SGM 2.0's “Community Engagement” domain recognises peer learning as a

teacher skill (MOE, 2023). PPGB 2.0 gives new teachers a structured induction and mentoring phase.

3. Shortcomings. IAB's training happens in Kota Kinabalu or Tuaran — six to ten hours from some Interior districts — so small schools cannot send their leader to multi-day courses. PKG visits take teachers away from already short-staffed classrooms. WhatsApp groups carry no recognition, no archive, no quality control, and vanish when a teacher transfers. There is no cross-school mentoring system after PPGB 2.0 induction ends. SSH 2023–2025 targets are set at state level, not at the cluster or school level where peer learning actually happens.

4. Proposed Solution Direction. A simple digital coordination layer — not a new platform — should let mentor teachers and #CikguJuaraDigital schedule short asynchronous check-ins with mentees across schools, share lesson materials in a searchable cluster folder, and log activity against SGM 2.0 “Community Engagement” indicators. It should work with the PKG cluster structure, not replace it. Guru Besar should see it as a dashboard without a new admin login. Recognition should attach to small concrete actions (one lesson shared, one mentoring check-in, one peer observation) instead of bulk attendance hours. Offline-first is required: materials made on a day with no internet must be submittable later from a NADI centre or the teacher's home town on weekend trips.

3.3 ContextSquaRE 3 — Project Goal 3

Project goal. District and national offices should be able to track, recognise, and scale up rural teacher training based on real evidence.

Stakeholders. BPG as national owner of training quality and the SGM 2.0 standard; JPN Sabah and the PPDs; BSTP as owner of DELIMa 2.0; IAB Cawangan Sabah for school leader results; MCMC for connectivity data; and Yayasan PETRONAS and UNICEF Malaysia as outside partners with their own tracking (Program Duta Guru and FS4A).

1. Key Challenges. On 30 December 2025, Minister Fadhlina Sidek announced that SPLKPM, BPG's central training tracking system, will be shut down in 2026 (Utusan Malaysia, 2025). This happens just as DPD 2023–2030 and RMK-13 start asking for more measured results. No public dataset breaks down teacher digital skill levels for Sabah Interior schools, and the 300 to 400 unconnected schools leave no trace in platform analytics (The Star, 2025). The data we do have is about attendance, not about what changed in the classroom. Rural teachers, being offline more often, leave less digital trace, so measurement is biased toward the already-connected. There is no shared dashboard between JPN Sabah and BPG for rural teachers as a group.

2. Existing Solutions. SGM 2.0 includes a self-assessment tool covering four skill domains (MOE, 2023). DELIMa 2.0 has some teacher-side

analytics for modules finished and micro-credentials earned (BSTP, 2025). SSH 2023–2025 has a four-pillar framework linked to TS25 (JPN Sabah, 2022). BPG promises yearly impact studies covering 100% of its programmes (MOE, n.d.-a). Outside partners track their own work: Program Duta Guru tracks progress toward 4,500 STEM teachers by 2030 (Yayasan PETRONAS, 2024); FS4A reports on reach and use in both online and offline teacher modes (UNICEF Malaysia, 2024).

3. Shortcomings. Attendance and hours are poor proxies for teaching change, which is one reason SPLKPM was shut down. SGM 2.0 self-assessment is not checked or combined across schools. DELIMa 2.0 analytics only cover online schools, excluding the ones that need help most. SSH targets sit at state level and do not break down to cluster or teacher. BPG's impact studies come out too late to change in-year spending. Outside-partner tracking is not connected to BPG's, so a teacher doing both Program Duta Guru and #CikguJuaraDigital is counted in their systems but not nationally. No tracking method exists for the rural teacher group that DPD 2023–2030 will be judged on.

4. Proposed Solution Direction. A light, offline-tolerant logging tool should support, not replace, the SGM 2.0 self-assessment. It should log teacher activity even at disconnected schools and sync when a connection is available. Recognition should link to SGM 2.0 skill domains (knowledge, instruction, community, and personal quality) rather than raw hours, so teachers do not carry the compliance burden that SPLKPM's shutdown was meant to remove. A district dashboard should serve JPN Sabah and the PPDs, while BPG gets a combined national view for DPD 2023–2030 and RMK-13 reporting. BPG should publish a mapping from SGM 2.0 to UNESCO ICT-CFT so the dashboard uses shared skill categories. The tool should be compatible with Program Duta Guru and FS4A from the start, so teacher work is counted once and visible everywhere.

4. Personas

In the section below, three personas are developed and each align with a single ContextSquaRE perspective. The personas are based on information we found about rural Sabah teachers and the institutional structures described in Section 2. The naming conventions are derived from Kadazandusun, Bajau, and Malay in Sabah Interior.

Persona 1: Cikgu Liana (rural primary school teacher) - Liana Anak Gunsalam is a 26 years old female and teaches Sekolah Kebangsaan, English and Pendidikan Moral. She obtained a Bachelor of Education (TESL), graduated from the IPG Kampus Keingau in 2023, and was posted to a Rural 2 school in Nabawan district in her second year of working. Liana owns an Android phone, a Redmi Note 12 and

a personal laptop that she bought with a SmartBelanja teacher loan. She uses a few apps on a daily basis, like: WhatsApp, TikTok, Shopee, Canva and Google Classroom. She does have a DELIMa account but she is only able to log in about once a month as the 4G of the school is quite unreliable. She has a Celcom 40GB internet card, but if does a 30-minute class, she spent all her GBs for that month. Liana is not married and lives at the school site because her family home is about 4.5 hours removed from the school. She has a DG41 base salary and also receives a BIMLTK Rural 2 allowance. The school has 87 students and 7 teachers in total. For all these students there is a single 4G router that mainly works in the morning and has frequent power cuts. Even though Liana flagged the internet issue to the State Education Minister, the problem has not been solved so far.

Problems

Liana wishes to receive more training to be able to provide better classes for the students, however the closest training location is in Kota Kinabalu, which is about 2 days of travel for 1 day of training. Online webinars on DELIMa are too heavy to run on the 4G connection she has available. The trainings available require accessories of modern classrooms, like tablets iPads, projectors and computers for each student, which are not available for Liana. Jaafar et al. (2025) described similar issues for rural teachers: weak infrastructure, a heavy workload and professional isolation.

Needs

Short lessons that do not require live presence, but are downloadable at a small memory and low bandwidth, so she can use the morning 4G at the school to do so. The lessons should be working offline as well, and easily transferable to paper if needed. The content should be in Bahasa Malaysia and include summaries for teachers in Kadazandusun to hand out to the student's parents. There should also be a way for her to keep track of her CPD record when she transfers.

Existing solutions

There are various small crumbs available, like watching YouTube videos when she is in Keningau town, or copy lesson ideas from Telegram/WhatsApp groups. There was a way to track CPD records, but unfortunately SPLKPM will be shut down in 2026 (Utasan Malaysia, 2025), so right now keeping track on paper is a solution.

Persona 2: Guru Besar Jainal (school leader) - Jainal bin Sairin is a 47-year-old male and serves as Guru Besar of a 96-pupil Rural 2 Sekolah Kebangsaan in Pitas district. He holds a Master of Education in Educational Management from UPSI (2019) and completed the NPQEL programme through IAB Cawangan Sabah in 2022 (IAB, 2025). He spent eleven years as a classroom teacher before

becoming Guru Besar in 2023. Jainal uses a school-issued laptop and his own Samsung Galaxy A54. He is confident with WhatsApp, Microsoft Word, Excel, and the JPN Sabah e-Operasi system, but less so with cloud collaboration tools. He manages five active WhatsApp groups covering staff, the PIBG, PPD Pitas, his NPQEL alumni network, and a kampung committee group for the school. Internet at school runs through a 4G modem that drops out during heavy rain. Jainal is married with three children studying in Kota Kinabalu, whom he sees on weekends. He leads 9 teachers, 4 of whom are in their first or second posting. His school is 2.5 hours from Kota Marudu and 6 hours from Kota Kinabalu on partly unpaved roads. He sits on the SSH 2023–2025 cluster committee for his zone (JPN Sabah, 2022) and reports monthly to PPD Pitas.

Problems

Jainal must release teachers for CPD without leaving classes uncovered, which is structurally impossible for a 9-teacher school. IAB courses in Kota Kinabalu or Tuaran cost him a teacher for a full week. PPGB 2.0 induction supports new teachers in their first year, but there is no structured mentoring framework after that period ends. Peer learning happens informally on WhatsApp but is unrecognised, unarchived, and disappears when teachers transfer. He is assessed on the SSH "Quality Teachers" pillar outcomes but has no school-level view linking his teachers' CPD activity to those targets. The shutdown of SPLKPM removes the only national log he previously consulted (Utusan Malaysia, 2025).

Needs

Jainal needs a lightweight coordination view, not a new platform, showing which of his teachers are actively engaged in peer learning across the cluster, what evidence they have submitted against SGM 2.0 domains, and which mentor relationships are active. A weekly summary by WhatsApp or email is enough; he does not need to log in daily. He wants recognition tied to small concrete actions (one lesson shared, one peer observation, one mentoring check-in) rather than attendance hours, and he needs the tool to be offline-tolerant and integrated with existing PKG cluster activity rather than running as a parallel structure.

Existing solutions

Jainal currently runs an informal mentoring system through WhatsApp, biweekly staff briefings, and occasional cluster meetings at PKG Pitas. He tracks teacher CPD hours in a personal Excel sheet and nominates one or two teachers per term for #CikguJuaraDigital events when MDEC visits the district (Sabah Media, 2024). Recognition for peer mentoring is verbal or through a paper certificate at the end of the school year.

Persona 3: Pegawai Norhaida (PPD district officer) - Norhaida binti Mohd Tahir is a 39-year-old

female serving as Penolong Pegawai Pendidikan Daerah (Sektor Pengurusan Akademik) at PPD Keningau, grade DG44. She holds a Master of Education in Curriculum from UMS (2020) and spent eight years as a secondary school teacher before moving to the district level six years ago. Norhaida is a confident user of Microsoft 365, Google Workspace, and the JPN Sabah reporting suite and built a small Power BI dashboard in 2024 for SSH reporting. She uses DELIMa 2.0 admin views weekly. She works mostly from a desktop in the PPD office in Keningau town, where fiber internet is stable, and visits schools 2 to 3 days a week with a laptop and 4G dongle. Norhaida reports to PPD Keningau and, through SSH 2023–2025, to JPN Sabah under Raisin bin Saidin (JPN Sabah, 2022). She oversees 64 primary and secondary schools in the Interior, of which she estimates that around a quarter still have weak or no internet access, consistent with the 300 to 400 figure for Sabah as a whole (The Star, 2025). She is the contact point for visiting partners including MDEC #CikguJuaraDigital, Yayasan PETRONAS Duta Guru, and UNICEF FS4A, and is expected to feed district-level data into BPG's yearly impact studies (MOE, n.d.-a).

Problems

The shutdown of SPLKPM (Utusan Malaysia, 2025) removes her main central log just as DPD 2023–2030 and RMK-13 begin demanding measured outcomes. SGM 2.0 self-assessment data is not aggregated above the school level. DELIMa 2.0 analytics exclude offline schools, exactly the schools she most needs evidence for. Data from outside partners such as Duta Guru, FS4A, and #CikguJuaraDigital arrive in different formats and count the same teachers separately across systems. She spends roughly two days a month manually reconciling spreadsheets for cluster reporting.

Needs

Norhaida needs a district dashboard that combines SGM 2.0 self-assessment data, evidence-based teacher activity (lessons shared, peer observations, mentor check-ins), and partner-program participation, with offline-tolerant data capture so that disconnected schools are not invisible in the data. Recognition should be tied to SGM 2.0 skill domains rather than raw hours, and mapped to the UNESCO ICT Competency Framework for Teachers (UNESCO, 2018) for international comparability. She also needs read-only views she can share with school leaders like Guru Besar Jainal, and an export format that BPG can roll up nationally for DPD 2023–2030 reporting.

Existing solutions

Norhaida currently pulls reports manually from DELIMa 2.0, e-Operasi, and partner Excel sheets into her own Power BI workbook. She visits NADI centres to verify activity at offline schools (Borneo

Post, 2025a) and relies on Guru Besar self-reports validated by school visits for yearly SSH reporting.

5. Assumptions for Requirement Analysis

Assumption 1: teachers in Sabah Interior have their own phones (Android), however they cannot use 4G throughout the day, or at a high bandwidth. Often the internet connection only works in the mornings.

Assumption 2: the Bahagian Profesionalisme Guru (BPG) will use the SGM 2.0 because the SPLKPM will not continue in 2026 so there is no other way to log CPD records (Utusan Malaysia, 2025; Ministry of Education Malaysia, 2023).

Assumption 3: we assume that the solution will not replace DELIMA, PPGB, #CikguJuaraDigital, Program Duta Guru, and UNICEF FS4A, but it should integrate with them. There are already substantial investments made in these programs, which we do not want to become unused.

Assumption 4: we also assumed that on average a teacher stays about 2-3 years at a Sabah Interior school before transferring. We could not find any explicit district-level tenure statistics published online.

Assumption 5: there are more than thirty languages in Sabah, so we used a default language Bahasa Malaysia. Kadazundusun, Iban and English were added for some of the modules.

6. Requirements for our Digital Solution

6.1 Functional requirements

Functional requirement 1: the solution needs to have an option with small modules for CPD that can be used on smartphones. They should be able to be downloaded so it can later be watched without access to the internet.

Functional requirement 2: the solution focuses more on processing images rather than videos because of the lower bandwidth internet availability. It should be easy for Liana to upload images that account for her classroom evidence, e.g., for SGM 2.0 skill domains.

Functional requirement 3: the solution has a smart option where you can queue your submissions and keep them loaded on the app. Until there is a steady internet connection, the submissions will be uploaded online.

Functional requirement 4: the solution should have a mentoring dashboard where mentors and mentees can see their check-ins and search for lessons in their own folders. This will help Jainal.

Functional requirement 5: the solution is able to provide the option to log shared lessons, peer observations, mentoring check-ins, and the submitted evidence. This is used for the SGM 2.0 PPB.

Functional requirement 6: for Guru Besar it is important to receive a summary view for the entire school without having to login, for example it can be sent to his WhatsApp/Telegram on a weekly basis.

Functional requirement 7: the solution also provides a dashboard for the district for PDD/JPN Sabah for SGM 2.0. The self-assessments, recognition events, and other program participation can be shown here. This is important for Norhaida.

Functional requirement 8: the solution should be available in different languages, depending on the availability of all 30 languages, we can add a variety of languages. The user just chooses their language once (because switching languages will use internet connection).

Functional requirement 9: the solution solves the issue for the CPD log recordings. It will be possible to log everything using the solution.

6.2 Non-functional requirements

NFR1 (Connectivity): the mobile client must work fully offline for at least 14 days between syncs. A complete CPD micro-module bundle must not exceed 25 MB so that teachers like Liana can download it during the morning window when 4G is available at school.

NFR2 (Device compatibility): the mobile client must run on Android 11 or higher with no more than 2 GB RAM and a maximum installed size of 200 MB. This reflects that most rural Sabah teachers own mid-range Android phones rather than high-end devices.

NFR3 (Performance): any screen in the app must load within 3 seconds on a Redmi Note 12-class device, regardless of whether the device is online or offline at that moment.

NFR4 (Data cost): a typical month of use, covering four modules and a set of evidence submissions, must consume no more than 200 MB of mobile data. This keeps the solution within reach of teachers on a 40 GB monthly SIM card who cannot afford to spend all their data on one platform.

NFR5 (Localisation): the interface must be available in Bahasa Malaysia and English from launch. Kadazandusun and Iban strings must be addable without changing the underlying code, so that new

language support can be rolled out as content becomes available.

NFR6 (Accessibility): the interface must meet WCAG 2.1 AA standards for colour contrast and tap target size. Parent-facing audio summaries must include narration to support lower-literacy users.

NFR7 (Security and privacy): all teacher and pupil data must be stored in line with Malaysia's Personal Data Protection Act 2010, as amended in 2024. Pupil faces in evidence photos must be blurred automatically on the device before upload, so that no identifiable images leave the school.

NFR8 (Interoperability): the system must expose REST APIs aligned to SGM 2.0 and UNESCO ICT-CFT v3 so that DELIMa 2.0, Program Duta Guru, and UNICEF FS4A can pull or push data without manual reconciliation by Norhaida or her colleagues.

NFR9 (Reliability): syncing must be idempotent, meaning that re-uploading a queued evidence item never creates a duplicate entry in the district dashboard. This matters most for teachers who submit from NADI centres on weekend trips after accumulating a backlog.

NFR10 (Maintainability): no school-side administrator role should be needed. All admin functions must sit at PPD level or above, so that Guru Besar like Jainal are not burdened with platform management on top of their existing workload.

NFR11 (Sustainability): hosting must use federal cloud procurement such as MyGovCloud or an equivalent, so that the per-school operating cost is zero and the solution does not collapse when project funding ends.

NFR12 (Auditability): all SGM 2.0 recognition events must be timestamped, attributable to a specific teacher, and exportable in a format that replaces the audit function previously handled by SPLKPM.

6.3 MoSCoW prioritisation

Must have: FR1, FR2, FR3, FR5, FR9, NFR1, NFR2, NFR3, NFR4, NFR7, NFR9, and NFR12. Together these form the minimum viable loop: a teacher at a disconnected school can access training content, submit evidence offline, and have it recognised against SGM 2.0 once a connection is available. Without this core, Liana cannot benefit from the solution at all, and the audit gap left by the shutdown of SPLKPM in 2026 remains open.

Should have: FR4, FR6, FR7, FR8 limited to DELIMa 2.0 and Program Duta Guru, NFR5 for Kadazandusun in selected modules, NFR6, NFR8, NFR10, and NFR11. These requirements make the solution useful for Jainal and Norhaida and connect it to the existing programmes described in Assumption 3. They are not strictly needed for an individual teacher to complete a module, but without them the school leader and district officer perspectives from ContextSquaREs 2 and 3 receive little value and the solution risks running as a parallel structure next to DELIMa 2.0 rather than integrating with it.

Could have: FR8 extended to UNICEF FS4A, and NFR5 extended to Iban. These add real value but depend on conditions that are not guaranteed for a first release: partner agreements with UNICEF Malaysia, and enough Iban-speaking Champion Teachers available through the #CikguJuaraDigital network to co-author content.

Won't have in this release: a pupil-facing client, parent-facing content beyond audio summaries, integration with KDCA cultural-content authoring tools, and any feature that requires a continuous school-side internet connection. These are excluded to keep the first release focused on the conditions Liana actually faces, as documented by the Sabah State Minister of Education in December 2025.

7. Conclusion

Rural Sabah's combination of disconnected schools, high teacher turnover, and the imminent shutdown of SPLKPM creates a situation in which a digital CPD solution is both necessary and timely. This report set out to identify the requirements for such a solution, grounded in the conditions that teachers, school leaders, and district officers actually face. Starting from BPG's strategic goal and working through three ContextSquaREs and three personas, the requirements in Section 6 address three connected problems: teachers at disconnected schools cannot access or complete CPD in any meaningful way, peer learning and mentoring at school level is happening but leaves no record and disappears when teachers transfer, and district and national offices are losing their only tracking system just as new policy targets come into force.

The must-have requirements reflect this directly. An offline-first mobile client with downloadable micro-modules, image-based evidence submission, and a queue-and-sync mechanism gives Liana a way to complete and document CPD without relying on a stable connection. A portable CPD record tied to SGM 2.0 skill domains means her progress follows her when she transfers, which matters in a context

where the average posting lasts two to three years. The auditability requirement closes the gap left by the shutdown of SPLKPM in 2026, without placing a new compliance burden on teachers.

The should-have requirements extend this to Jainal and Norhaida. A lightweight coordination layer for peer learning and mentoring, a weekly school summary pushed to WhatsApp, and a district dashboard that aggregates SGM 2.0 data across online and offline schools make the solution useful beyond the individual teacher level. Integration with DELIMa 2.0 and Program Duta Guru ensures the solution fits into existing infrastructure rather than duplicating it, which is important given the investments already made in those programmes.

The could-have and won't-have categories reflect honest constraints. A national BPG rollup mapped to UNESCO ICT-CFT and full FS4A integration are worth building eventually, but depend on partner agreements and a published SGM 2.0 to ICT-CFT mapping that does not yet exist. A pupil-facing client and anything requiring continuous school connectivity are out of scope for this release entirely, in line with the conditions reported by the Sabah State Minister of Education in December 2025 (The Star, 2025).

Taken together, the requirements describe a solution that is narrow enough to be buildable under real rural constraints, and connected enough to the existing policy landscape to be adopted and maintained after the first release.

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